

Synthetic DCE from Non-Contrast Prostate MRI

Methods and Results — UCSF cohort, Tier-1 static track

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Abstract

We predict a post-contrast DCE volume from three non-contrast sequences (T2w, DWI, ADC) across 3,667 UCSF prostate exams, motivated by gadolinium avoidance in GFR-contraindicated patients and by cross-protocol harmonization. A 3D conditional GAN attains within-gland co-localization $r = 0.482$ against -0.043 for the strongest trivial baseline, establishing that the model predicts enhancement structure rather than passing anatomy through. The central result is a decomposition: within-gland *structure* is predictable, per-patient enhancement *amplitude* is not — and amplitude carries 51% of the signal. That deficit is then largely repaired by supplying the pre-contrast phase as a fourth input channel, which requires no gadolinium: brightness tracking rises from $r = 0.334$ to 0.76 and the model clears a pre-contrast passthrough control by $+0.267$ *roi_pearson*. Amplitude is therefore unidentifiable from bpMRI alone rather than unidentifiable in principle. Methods and the measurements that justify them are presented together throughout.

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1 Task and notation

Let $x = (x^{\text{T2w}}, x^{\text{DWI}}, x^{\text{ADC}}) \in \mathbb{R}^{3 \times D \times H \times W}$ be the stacked non-contrast input and $y \in [-1, 1]^{1 \times D \times H \times W}$ the target DCE phase. We learn $G_\theta : x \mapsto \hat{y}$, $\hat{y} \in [-1, 1]$ via tanh. $\mathcal{M}_i$ is the prostate mask of case i , occupying $\approx 5\%$ of the crop; Ω is the full volume. All intensities live in $[-1, 1]$, so SSIM and PSNR use data range $L = 2$.

2 Data

2.1 Source and cohort construction

Single-center UCSF, delivered across two secured mounts: anatomy and masks on DS2 (5,962 patients), 4D DCE with per-phase timing on DS3. All volumes arrive pre-registered into T2w space. Filters are applied at two points — the enhancement criterion at staging, so the tree is correct by construction, and two load-time flags.

Stage	Criterion	Remaining
Patients with DCE	—	5,439
Successfully staged	inputs resolvable, mask matches DCE grid	4,975
Enhancement QC (at staging)	$\max_t \text{enh}(t) \geq 1.5$	4,246
DWI b-value filter	DWI resolves to $b \geq 600$	3,684
Enhancement QC recorded	—	3,667
Patient-level split (train / val / test)		2805 / 312 / 550

Two attrition steps are findings, not bookkeeping. Roughly 12–13% of patients never enhance at *any* timepoint. Enhancement QC keys on the maximum over the whole series rather than the selected phase, because only the maximum separates a genuinely non-enhancing study (drop) from a mis-selected phase (recoverable). **Caveat:** this is a mask-mean ratio, so a mis-registered mask depresses it exactly like a failed injection; we now capture per-case registration quality but have not yet correlated the two, so the “failed injection” reading is not established.

Separately, 13% of DWI resolves to $b=50$, which carries almost no diffusion weighting and reads T2-like — a *different contrast* occupying the same input channel as the $b=1000$ the majority receive. Excluding it costs 13% of the data and is a clean one-flag ablation. A further 43 cases use *computed* (`_synth_`) high- b images rather than acquired ones.

2.2 Target phase: selected by time, not index

Enhancement relative to pre-contrast is

$$\text{enh}(t) = \frac{\mathbb{E}_{v \in \mathcal{M}}[S_v(t)]}{\mathbb{E}_{v \in \mathcal{M}}[S_v(0)]}. \quad (1)$$

Measured: flat at ≈ 1.0 until $t \approx 45$ s; wash-in 55–90 s ($1.12 \rightarrow 2.00$); then a **plateau** of 2.1–2.5 from ≈ 90 s out past 300 s. There is no sharp peak, so $\arg \max_t \text{enh}(t)$ lands on plateau noise at ≈ 280 s, drifting into washout.

Protocols are heterogeneous — 26–70 phases at 1.7–13 s cadence — so a fixed *index* is a different physiological moment per patient. We therefore select by time,

$$j^* = \arg \min_j |t_j - t_{\text{target}}|, \quad t_{\text{target}} = 120 \text{ s}, \quad (2)$$

inside the stable plateau, past the timing-sensitive wash-in, at $\approx 92\%$ of maximum. Realized selection times cluster tightly at 104–125 s.

Two protocol complications, handled explicitly. *Interleaved series* (27 cases): some studies stack k volumes per timepoint (Dixon water/fat), so $n_{\text{phases}} = k \cdot n_{\text{timepoints}}$ and a plain index alternates between sub-series, one of which barely enhances. Detected via repeated timestamps; the phase is re-picked within the timepoint, each sub-series scored against its own $t=0$. *Corrupt timing* (155 cases): some timing files carry impossible trailing values (e.g. 67989 s). Only the clean monotonic prefix is trusted, and where the time-selected phase still looks weak the phase is re-selected from the enhancement curve at plateau onset (first phase $\geq 0.9 \text{ max}$).

2.3 Staging

Each 4D series is ≈ 1 GB; training directly against them re-reads the full 4D per sample per epoch and depends on a mount that has historically dropped on GPU nodes. Staging reads each 4D *once*, writes the selected phase as 3D, and co-locates every modality in a flat per-patient directory with a metadata sidecar. The staged target is verified bit-identical to the raw-4D path. Current tree: 463 GB over 4,251 cases.

3 Preprocessing

3.1 Geometry: a fixed field of view, not a fixed pixel count

Measurement first. UCSF in-plane resolution is not uniform — matrices of 256/512/672/704/736/1024 at 0.176–0.781 mm. A fixed *pixel* crop of 256 therefore covers a wildly variable physical extent (full cohort, $n = 4246$):

256-px crop covers	Cases	Share
< 80 mm	734	17%
80–110 mm (intended)	3346	79%
> 110 mm	164	4%

An $8\times$ spread in anatomical scale (40–320 mm), with 172 cases cropped to 45 mm — narrower than a prostate, so the crop sits *inside* the gland, the mask fills the frame, and ROI metrics

degenerate to whole-image metrics. The through-plane axis has the same defect in miniature: 97.8% of cases are 3.00 mm slices, but 44 are 0.70–0.80 mm, giving those a 25.6 mm slab where everyone else gets 96 mm.

Method. Resample all modalities to a common physical grid before cropping, (0.47, 0.47, 3.0) mm for 3D, then center-crop to $32 \times 192 \times 192$: a fixed $90 \times 90 \times 96$ mm field of view for every patient. Validated as a near-no-op for the dominant 512@0.352 mm group and the 97.8% at 3.00 mm, and T2w/DCE grids are identical in all 4,246 cases so the iso anchor introduces no crop-center shift.

3.2 Intensity normalization

Single center, so there is no cross-scanner distribution to align and no Nyul landmark fitting is used. DCE is robust-scaled about the body-tissue median m with spread s :

$$y \mapsto \text{clip}\left(\frac{y - m}{k s}, -1, 1\right), \quad k = 1. \quad (3)$$

Justification. Within-prostate target dispersion measured over 12 staged cases: $k=3 \Rightarrow 0.073$, $k=2 \Rightarrow 0.109$, $k=1 \Rightarrow \mathbf{0.218}$. Larger k flattens the target until `roi_pearson` and `ssim_roi` become noise on a near-uniform field. T2w uses percentile normalization, ADC a fixed physical clip (preserving absolute mm^2/s), DWI a per-image foreground z -score.

3.3 Split

Patient-level, seeded, 15% held out; validation carved from the training split only. **Note:** `--seed` currently controls both the split and training initialization, so multi-seed repeats vary both unless a separate split seed is added. ID prefixes were tested as a possible batch marker and rejected (`8ii*` spans 512, 704, 736 and 1024 matrices alike).

4 Models and objective

Families. 3D conditional GAN (pix2pix-style U-Net generator with full skips, hinge loss, projection discriminator); 3D latent flow matching and DDPM sharing a frozen first stage — either a trained `AutoencoderKL3D` or the frozen MedVAE foundation VAE. Every trainer returns $(\text{model}, \text{gen})$ with $\text{gen}(x) \rightarrow \hat{y}$, so evaluation is model-agnostic. A 2D track (pix2pix, pixel-space CFM, MedVAE-latent flow) turns $\approx 3.2\text{k}$ volumes into $\approx 52\text{k}$ slices at $\approx 6\times$ lower cost per step.

Objective. With ϕ a frozen 3D MedicalNet feature extractor,

$$\mathcal{L} = \lambda_1 \mathcal{L}_1^{\text{ROI}} + \lambda_s (1 - \text{SSIM}_{3D}) + \lambda_s (1 - \text{SSIM}_{\mathcal{M}}) + \lambda_p \|\phi(\hat{y}) - \phi(y)\|_2^2, \quad (4)$$

$$\mathcal{L}_1^{\text{ROI}} = \frac{\sum_{v \in \Omega} w_v |\hat{y}_v - y_v|}{\sum_{v \in \Omega} w_v}, \quad w_v = 1 + (\omega - 1) \mathbb{I}[v \in \mathcal{M}], \quad \omega = 10. \quad (5)$$

Without ROI weighting the loss is dominated by background and padding. **Asymmetry worth stating:** the latent models train on latent-space MSE with no clean mapping to the mask, so ROI emphasis reaches them only through the first stage’s reconstruction. Any GAN-versus-flow comparison inherits this.

Flow matching. On the rectified path with $t=0$ at the data and $t=1$ at the source,

$$z_t = (1 - (1 - \sigma_{\min})t)z_0 + t\epsilon, \quad v^* = \epsilon - (1 - \sigma_{\min})z_0, \quad \epsilon \sim \mathcal{N}(0, I), \quad (6)$$

$\sigma_{\min} = 0$, $t \sim \mathcal{U}(0, 1)$ per example. The clean-latent identity $\hat{z}_0 = z_t - t v_\theta(z_t, t)$ holds exactly for any $\sigma_{\min}$ and is what the optional image-space anchor decodes and supervises.

5 Metrics

Everything ROI is per-case, then averaged. Pooling masked voxels across the batch mixes between-patient brightness into every statistic. On synthetic data where per-patient brightness is predicted perfectly and within-gland structure is pure noise, pooling reports $r = +0.977$ where the truth is $+0.020$, and the value climbs with batch size (0.66 at $B=2$, 0.98 at $B=8$). This was a live bug; it is now pinned by assertion.

$$r_i = \frac{\sum_{v \in \mathcal{M}_i} (\hat{y}_v - \bar{\hat{y}})(y_v - \bar{y})}{\sqrt{\sum_{v \in \mathcal{M}_i} (\hat{y}_v - \bar{\hat{y}})^2} \sqrt{\sum_{v \in \mathcal{M}_i} (y_v - \bar{y})^2}}, \quad \text{roi_pearson} = \frac{1}{N} \sum_i r_i, \quad (7)$$

$$\text{roi_var_ratio} = \frac{1}{N} \sum_i \min\left(\frac{\text{Var}_{v \in \mathcal{M}_i}[\hat{y}]}{\text{Var}_{v \in \mathcal{M}_i}[y] + \varepsilon}, 5\right), \quad \text{roi_grad_ratio} = \frac{1}{N} \sum_i \min\left(\frac{\mathbb{E}_{v \in \mathcal{M}_i} \|\nabla \hat{y}\|}{\mathbb{E}_{v \in \mathcal{M}_i} \|\nabla y\| + \varepsilon}, 5\right), \quad (8)$$

both with target 1 (< 1 over-smoothed, > 1 over-textured). The histogram distance uses order statistics, $\text{roi_w1} = \frac{1}{2} |\mathcal{M}_i|^{-1} \sum_k |\hat{y}_{(k)} - y_{(k)}|$, and a label-free composite

$$\text{realism} = \text{mean}\{\nu(\text{var_ratio}), \nu(\text{grad_ratio}), 1 - \min(\text{w1}, 1)\}, \quad \nu(x) = 1 - \min(|x - 1|, 1). \quad (9)$$

The composite is a ranking heuristic, not a validated metric — equal weights, no weighting study. Report components individually. **The $\min(\cdot, 5)$ caps bias both ratios** toward 5 from above; they exist because a near-flat target denominator sent uncapped values into the thousands.

Global SSIM/PSNR are $\approx 99\%$ background and near-meaningless here. The *gap* between global and ROI values is the diagnostic.

6 Results

6.1 Model comparison

3D models, best checkpoint, held-out test ($n = 550$). VAL $\approx$ TEST throughout.

	3D GAN	3D flow (MedVAE)	GAN + pre-contrast
roi_pearson $\uparrow$	0.482	0.369	0.527
ssim_roi $\uparrow$	0.397	0.276	0.449
MAE $_{\mathcal{M}}$ $\downarrow$	0.236	0.269	0.200
roi_var_ratio $\rightarrow 1$	0.912	1.711	0.951
roi_grad_ratio $\rightarrow 1$	0.711	1.059	0.773
p75_corr $\uparrow$	0.357	0.351	0.743
realism $\uparrow$	0.845	0.714	0.887
FID $\downarrow$	88.9 †	169.8 †	84.6
PZ / TZ roi_pearson	0.339 / 0.522	0.169 / 0.417	0.445 / 0.547

[†] measured on the *last* checkpoint, not the best — see §7.1.

The GAN leads the flow on faithfulness across four independent setups (2D and 3D, pixel and latent). The flow’s one advantage is texture (`grad_ratio` = 1.06 vs 0.71): it is less smooth, which reads as more realistic while localizing worse. **Two confounds:** checkpoints are selected on `ssim_roi`, which is smoothness-biased and therefore favours the GAN’s failure mode; and the flow’s objective is effectively ROI-unweighted (§4). A selection-metric ablation has not been run.

The 2D ranking reverses under pre-contrast conditioning. With the fourth channel the 2D pixel-space flow reaches `roi_pearson` = 0.439 against the 2D GAN’s 0.293 — the opposite of every prior comparison — and the 3D flow gains more than the 3D GAN (+0.097 vs +0.058; PZ +0.148 vs +0.112), halving the 3D gap from 0.113 to 0.066. This is consistent with the flow’s characteristic failure having been amplitude (over-dispersion, and worse than a constant predictor on MAE), which is exactly what pre-contrast repairs. Both 2D numbers are $n = 1$ and there is no 2D noise floor yet, so the reversal is not established — but “the GAN wins” can no longer be stated as a general result.

Peripheral zone. $r_{\text{PZ}} = 0.339$ against $r_{\text{TZ}} = 0.522$ for the base GAN. PZ is where most clinically significant cancer arises and where DCE drives the PI-RADS upgrade rule, so this is the clinically weakest region. Pre-contrast conditioning lifts PZ disproportionately (+31% against TZ’s +5%).

6.2 The identifiability decomposition

Three predictors requiring no learning, with $\mu_i = \mathbb{E}_{v \in \mathcal{M}_i}[y_v]$, $\bar{\mu} = \mathbb{E}_i[\mu_i]$: $\hat{y}_v^{\text{const}} = \bar{\mu}$; $\hat{y}_v^{\text{level}} = \mu_i$ (an *oracle* knowing correct per-case brightness with no structure); $\hat{y}_v^{\text{t2w}} = x_v^{\text{T2w}}$.

Predictor	MAE _{$\mathcal{M}$}	<code>roi_pearson</code>	Knows
<code>const</code>	0.2510	—	nothing
<code>t2w copy</code>	0.3372	−0.043	identity baseline
MODEL	0.2355	+0.482	trained generator
<code>level (oracle)</code>	0.1851	—	brightness only, no structure

Total ROI variance decomposes exactly, by the law of total variance:

$$\text{Var}_{i,v}[y] = \underbrace{\text{Var}_i[\mu_i]}_{\text{amplitude}} + \underbrace{\mathbb{E}_i[\sigma_i^2]}_{\text{structure}}, \quad \sigma_i^2 = \text{Var}_{v \in \mathcal{M}_i}[y_v]. \quad (10)$$

Measured: $\text{sd}_i[\mu_i] = 0.231$, $\mathbb{E}_i[\sigma_i] = 0.226$, giving an amplitude share of $0.231^2 / (0.231^2 + 0.226^2) = \mathbf{0.51}$. **This is an upper bound:** we substituted $(\mathbb{E}[\sigma])^2$ for $\mathbb{E}[\sigma^2]$, and Jensen makes the true structure term at least that large.

Reading it. *Structure is learned* — T2w-copy scores ≈ 0 , so $r = 0.482$ is genuine localization, not laundered anatomy. *Amplitude is not* — the model tracks per-patient brightness at $\text{corr}_i(\mu_i, \hat{\mu}_i) = 0.334$ and compresses between-case spread $2.4\times$ ($0.231 \rightarrow 0.098$). Because amplitude is half the variance, both models lose to the oracle-brightness baseline in MAE, and the flow is worse than a *constant* predictor (−0.018) while retaining real localization ($r = 0.369$).

Why this is the task, not the model. Per-case amplitude depends on gadolinium dose, injection rate, cardiac output, renal clearance and receiver gain — none visible in T2w/DWI/ADC. We inspected the source DICOM headers for exactly these variables: they were removed by de-identification (dose, rate, patient weight and field strength absent; manufacturer overwritten with "NA"; only patient age survives). The information is not merely unused — it is absent.

Revised below. §6.3 shows this deficit is largely repaired once the pre-contrast phase is available: the claim holds for bpMRI alone, not in general.

Why it may not matter clinically. PI-RADS curve types are scale-invariant: $\text{type}(\alpha C(t)) = \text{type}(C(t))$ for any $\alpha > 0$, since Type I (progressive), II (plateau) and III (washout) are defined by post-peak *shape*. The unidentifiable component is orthogonal to the endpoint.

6.3 Pre-contrast conditioning

Supplying the pre-contrast phase (DCE phase 0, same acquisition and geometry) as a fourth input channel, identical configuration otherwise:

	Base GAN	+ pre-contrast	Δ
roi_pearson	0.482	0.527	+0.045 ($\approx 2.6\times$ noise)
p75_corr (amplitude)	0.357	0.743	+108%
PZ roi_pearson	0.339	0.445	+31%
MAE _M	0.236	0.200	−15%
Global SSIM / PSNR	0.469 / 16.8	0.697 / 21.1	+49% / +4.3 dB

Passthrough control. Pre-contrast is a far stronger predictor of post-contrast than T2w — same acquisition, same receiver gain, post $\approx$ pre + enhancement — so a model handed that channel could depress MAE simply by copying it. Adding $\hat{y}_v = x_v^{\text{pre}}$ to the baseline panel settles it:

Predictor	MAE _M	roi_pearson	Knows
const	0.2510	—	nothing
t2w copy	0.3372	−0.043	identity baseline
PRE copy	0.2855	+0.260	pre-contrast passthrough
MODEL	0.2005	+0.527	trained generator
level (oracle)	0.1851	—	brightness only, no structure

The model clears passthrough by +0.267 **roi_pearson** ($\approx 16\times$ the noise floor) and −0.085 MAE. It is not copying: pre-contrast is used as a calibration reference *and* structure is added on top.

Amplitude is substantially recoverable — a correction. §6.2 argued that per-case amplitude is unidentifiable. That was too strong: it is unidentifiable *from bpMRI alone*. Every stable amplitude readout moves sharply once pre-contrast is available.

	Base GAN	+ pre-contrast	Target
brightness tracking $\text{corr}_i(\mu_i, \hat{\mu}_i)$	0.334	0.76	1.0
between-case predicted sd	0.098	0.189	0.231
bias sd	0.218	0.151	—
gap to oracle brightness (MAE)	+0.0505	+0.0154	≤ 0
improvement over const (MAE)	+0.0154	+0.0504	> 0

Spread compression falls from $2.4\times$ to $1.22\times$, and the model closes from 27% worse than the oracle-brightness baseline to 8% worse.

The two-component split survives; the sizes were wrong. Amplitude decomposes into an *acquisition/baseline* term (receiver gain, the tissue baseline the gland enhances from) and an *injection/physiology* term (dose, rate, cardiac output, renal clearance). Pre-contrast — same series, same gain — supplies the first, and it turns out to dominate. The residual ($r = 0.76$ rather than 1.0; spread still 18% compressed; MAE still short of the oracle) is plausibly the second, which de-identification removed from the headers and which no image contains. Clinically this costs nothing: a pre-contrast T_1 requires no gadolinium, so the premise of the work is intact.

Seed repeats. Four runs at seeds 0–3 with the split held fixed (`--split-seed 0`), against $n = 2$ without pre-contrast:

	no pre-contrast ($n=2$)	+ pre-contrast ($n=4$)
roi_pearson	0.482, 0.466	0.532 ± 0.024
ssim_roi	0.397, 0.389	0.453 ± 0.018
PZ roi_pearson	0.339, 0.330	0.447 ± 0.024
p75_corr	0.357, 0.081	0.748 ± 0.028

The groups do not overlap: the *worst* pre-contrast run (0.500) exceeds the *best* run without (0.482). PZ separates by 0.112, $\approx 4.7\times$ the standard deviation. The measured $\text{sd} = 0.024$ also revises the earlier noise estimate upward from 0.017, which was extrapolated from a single replicate pair.

7 Reliability

7.1 Pipeline corrections

Every defect below was *silent* — no crash, no warning, plausible-looking output. Grouped by remediation cost: measurement defects can be re-scored from checkpoints, training defects need retraining, data defects need re-staging.

Defect	Effect	Evidence
Pooled ROI statistics	cross-patient brightness masqueraded as within-gland correlation	$r = +0.977$ on pure noise where truth is $+0.020$
Transposed convolutions	period-2 checkerboard stamped into every prediction	lag-1 autocorr. -0.60 vs $+0.81$ real; FID $175 \rightarrow 89$
Pixel-count crop	$8\times$ range of anatomical scale	40–320 mm across cohort
Final-epoch evaluation	3D <code>metrics.json</code> scored a degraded model	roi_pearson $0.419 \rightarrow 0.482$
Corrupt-timing fallback	155 cases trained against the <i>pre-contrast</i> volume	re-selected at plateau onset
Re-stage without QC	729 pruned non-enhancers silently resurrected	a run trained on 4,975
Harmonizer leakage	Nyul fit on the full cohort incl. test	never triggered by our configs

The checkerboard detector is the lag-1 autocorrelation of the first difference, $\rho = \text{corr}(d_k, d_{k+1})$ with $d_k = y_{k+1} - y_k$, which tends to -1 under a pure checkerboard and -0.5 for white noise. Predictions scored **-0.598** — alternating in sign at pixel scale *more strongly than white noise* — against $+0.813$ for real DCE, with $5\times$ the target’s high-frequency power. Post-fix: $+0.278$ and HF power at the target’s level.

A 28-assertion preflight (`python -m tier1_static.selfcheck`) pins each of these. The remaining leakage surface was audited clean: train/test disjoint and exhaustive; val carved from train only; `test_loader` never used during training; latent scaling computed from train only; 2D splits patients before slicing; evaluation and validation share clamping and metric code.

7.2 Run-to-run variance

Two *configuration-identical* runs at the same seed differ materially. Seeding covers Python/NumPy/Torch but not cuDNN autotuning or non-deterministic `atomicAdd` in 3D convolution backward; adversarial dynamics amplify the divergence; and best-checkpoint selection converts a bumpy validation curve into a noisy discrete choice of epoch.

Metric	Δ between identical runs
<code>roi_pearson</code>	0.017
<code>ssim_roi</code>	0.008
<code>roi_var_ratio</code>	0.055
<code>p75_corr</code>	0.276

Rule. Differences below ≈ 0.02 `roi_pearson` are unresolved at $n = 1$. *Safe:* GAN > flow (0.113, $\approx 7\times$); model $\gg$ t2w ($\approx 30\times$); PZ < TZ (paired within run); best > last checkpoint ($\approx 4\times$). *Not safe:* any `p75_corr` comparison at $n = 1$; the iso-versus-no-iso A/B (0.020 — inside noise, so iso is justified on correctness grounds only). This is one replicate pair, bounding the magnitude but not a standard deviation; ≥ 3 seeds are required.

A prediction that held: instability is a property of the target, not the metric. `p75_corr` swung 0.276 between two *identical* runs without pre-contrast. Across four pre-contrast runs it is ± 0.028 — a tenfold reduction, same metric, same code, same cohort. The only change is that the quantity it measures became predictable. This is the sharpest available evidence for the claim below: the metric was unstable *because* its target was not determined by the inputs, and supplying the missing input removed the instability. It was predicted before the pre-contrast run and is therefore a falsifiable test rather than a post-hoc reading.

Why `p75_corr` was $16\times$ worse. `roi_pearson` averages $N = 550$ per-case values, suppressing variance by $\sqrt{N}$. `p75_corr` is a single cross-patient correlation measuring the *amplitude* dimension — precisely what nothing in the loss constrains. **Reproducibility tracks identifiability:** the well-determined component reproduces, the ill-determined one drifts. The instability is a measurement of which part of the task is ill-posed, not merely a nuisance.

8 Tier-2: kinetics

The multi-timepoint target is the per-voxel enhancement curve. Unlike generic video world models, the governing dynamics are *known* — the Tofts model,

$$\frac{dC_t}{dt} = K^{\text{trans}}C_p(t) - k_{ep}C_t(t), \quad k_{ep} = K^{\text{trans}}/v_e, \quad (11)$$

with C_p the arterial input function. A continuous-time latent world model integrates a learned velocity field in physical time,

$$\frac{dz_t}{dt} = v_\theta(z_t, t; z_0, c) \implies z_T = z_0 + \int_0^T v_\theta dt, \quad (12)$$

with z_0 the encoded pre-contrast state and c a conditioning signal. The proposed contribution is to *constrain* v_θ to the Tofts family with parameter maps predicted from bpMRI rather than learn it free-form — a stronger inductive bias with guaranteed physically plausible curves. This directly addresses the limitation such methods cite, that physics-informed ODE approaches “require explicit prior knowledge of the true governing equations”; for DCE we have it.

Two obstacles to state up front. Tofts governs *concentration*; we have normalized signal, and converting requires a baseline T_1 map this cohort lacks, so recovered parameters are semi-quantitative at best. And an arterial input function is required, which a 90 mm prostate-centred crop may not contain.

Goal state. Acquisition duration (measured, $n = 1367$): $p_5 = 222$ s, $p_{50} = 300$ s, $p_{95} = 380$ s. Coverage: 99.0% at 180 s, **91.9%** at 240 s, 49.7% at 300 s. We anchor at 240 s — past plateau onset by ≈ 150 s, inside the window where Types II and III separate, retaining 92% of the cohort. The 8% falling short run the same protocol at faster cadence (median 45 phases versus 36), not a different scanner, so excluding them from a curve-type readout introduces no protocol confound.

9 Paper partitions and venues

	Claim	Still needs	Venue
1	Single-phase synthetic DCE is clinically usable	labels; reader study; multi-seed	Radiology / Lancet Dig. Health
2	Standard synthesis metrics do not measure case-specific fidelity	within-model sweep; ≥ 3 seeds; <i>second domain</i>	ICLR (tight) $\rightarrow$ ICML / NeurIPS
3	A physics-constrained velocity field beats a free-form one	Tofts parameterization; interpolation ablation	NeurIPS workshop $\rightarrow$ MICCAI

Track 2 is the strongest near-term result and is *label-free*: the decomposition of §6.2, the reproducibility-tracks-identifiability observation, and now a constructive counterpart — the missing component is recovered by naming the right input, which turns “here is something that cannot be predicted” into “here is which input carries which component of the target,” a stronger and more useful claim. Its framing must avoid the perception–distortion tradeoff, which is established; the contribution is a *diagnostic for whether a conditional model used its conditioning at all*. A single medical domain reads as an application paper, so a second domain is the gating requirement.

Track 3’s novelty is not porting a world model to a new organ — that is a MICCAI paper — but that we know the governing equation.

On the Tier-1 $\rightarrow$ Tier-2 gate. The project’s onboarding sets a hard gate at PI-RADS $\kappa \geq 0.6$. That is currently *unmeasurable* rather than failed: the overall PI-RADS category is essentially absent from the report export (92/10,538 once a boilerplate footer URL is excluded), though the binary DCE component appears in $\approx 6,400$ reports. Two developments argue for revisiting it: the scarcity rationale has evaporated (UCSF supplies $\approx 8\times$ the planned Tier-2 pool at $\approx 10\times$ the temporal density), and curve types are scale-invariant, so the one component we cannot predict is orthogonal to the endpoint. Gating on *DCE-positivity* κ is the defensible substitute.

10 Anticipated technical questions

Is $r = 0.482$ inflated by zonal contrast rather than focal enhancement? A model learning only “TZ is brighter than PZ” would score well without detecting anything. The data partly answers it: if the between-zone term were doing the work, whole-gland r should greatly exceed both within-zone values. It does not — 0.482 sits *between* $r_{PZ} = 0.339$ and $r_{TZ} = 0.522$, and both within-zone values are solidly positive. This does *not* establish lesion-level detection, which needs lesion labels.

You report a mean over 550 cases with no interval. Correct, and it should be added — the per-case r_i distribution and a bootstrap CI are computable from data in hand. The ± 0.017 figure bounds *run* variability, not *case* variability; the two must not be conflated.

Why Pearson? The signal–concentration relationship is nonlinear. A real limitation. MR signal relates to concentration through T_1 relaxation and the spoiled gradient-echo signal equation. Spearman ρ is the rank-based alternative and is trivial to add; reporting both would be more defensible.

Does the variance decomposition assume independence? No — the law of total variance is an identity. The only caveat is the $(\mathbb{E}[\sigma])^2$ -for- $\mathbb{E}[\sigma^2]$ substitution, making 51% an upper bound.

Is “the GAN wins” a checkpoint-selection artifact? Partly. `ssim_roi` is smoothness-biased and the GAN is the smoother model. Against that: the GAN also leads on `roi_pearson`, which is not smoothness-biased, and by $\approx 7\times$ the noise. A selection-metric ablation is the clean answer and is not done.

Were the 3D runs converged? 40 epochs versus 100 for 2D. A wall-clock choice ($\approx 670\text{s/epoch}$), not a convergence criterion, and convergence was not demonstrated. Against it: the GAN *degrades* late (discriminator wins, $d \rightarrow 0.03$), and the selected checkpoint clearly beats the final epoch. Per-run validation trajectories are now logged and will answer this directly.

Are the non-enhancers biology or registration failure? Not separated. A mis-registered mask depresses mask-mean enhancement exactly like a failed injection. Per-case registration quality is now captured (cohort median 0.459, $p_{90} = 1.694$) but has not been correlated against enhancement.

FID on 550 grayscale volumes with an ImageNet backbone? Weak. FID is biased at small n (KID is the alternative) and Inception features are natural-image. It is retained because it responded sharply and correctly to a known artifact ($175 \rightarrow 89$), but it should not carry a headline claim.

How were the masks produced? Automatic segmentations delivered with the cohort; none manually verified. Every ROI number here is conditioned on them.

Is the effective test sample really 550? Probably smaller. Protocols cluster (generic, Siemens TWIST, GE DISCO) and the split is not stratified by protocol. Stratifying — now possible, since the protocol string is captured — would fix this and double as the multi-vendor generalization result.

11 Reproducing this

Cohort. Stage once with the QC filter inline — applying it at staging rather than pruning afterwards is what makes the tree correct by construction:

```
MIN_ENH=1.5 sbatch --array=0-3 scripts/stage_ucsf.slurm      # -> 4,246
python -m tier1_static.data.stage_ucsf --report             # verify before training
python -m tier1_static.selfcheck                             # 28 assertions, CPU, seconds
```

Expect median enhancement ≈ 2.07 , $p_{10} = 1.70$, 27 interleaved, 155 curve-reselected, and no $t=0$ warning. Training logs must show min.b=600 and 562 below b600, 17 without enhancement QC.

Training.

```
UCSF_ROOT=<staged> MODEL=gan OUTPUT_DIR=runs/<name> \
REFERENCE=iso ISO_SPACING="0.47 0.47 3.0" SPATIAL_SIZE="32 192 192" \
T2W_NORM=percentile DCE_NORM=robust DCE_ROBUST_K=1 \
EPOCHS=40 BATCH_SIZE=4 BASE_CH=32 ROI_WEIGHT=10 VAL_EVERY=2 SEED=0 \
EXTRA_ARGS="--dwi-min-bvalue 600 --require-qc" sbatch scripts/train.slurm
```

The flags must match between training, evaluation and audit, or the checkpoint is scored on a different distribution than it was trained on. `config.json` is written by both the 2D and 3D paths for exactly this reason. Each run also emits `val_trajectory.jsonl`, one row per validation step with the full metric dict — the faithfulness-versus-realism trajectory of §7.2 is read from there.

Baseline audit. Same flags plus `--audit-split test`:

```
python -m tier1_static.audit_baselines --model gan ... --audit-split test
```

Multi-timepoint staging. Anchors on a common time grid, dense through wash-in and sparse across the plateau, since that is where curve types separate:

```
--anchor-times 45 60 75 90 150 180 240      # 0s and 120s already staged
```

Seven additional phases cost ≈ 0.76 TB over the existing 463 GB, and one read of each 4D rather than seven.

12 Limitations

2D and 3D ROI metrics are not comparable (the 2D dataset keeps only gland-containing slices, so denominators differ). FID for the base models was measured on the last checkpoint. Every result predating the audit is superseded. No claim is made that more data would close the amplitude gap: an earlier conclusion to that effect was measured on a defective configuration and has been retracted. The pre-contrast result awaits its passthrough control.